

Bilal Ahmed

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SUMMARY

Computer Science student at NUST with hands-on experience building AI-powered full-stack applications using LangChain, FastAPI, and React. Developed FinMate AI, a RAG-based banking assistant leveraging ChromaDB and Groq (LLaMA3) for grounded, citation-backed responses. Focused on practical AI integration and building clean, deployable code.

EDUCATION

National University of Sciences and Technology (NUST) Islamabad, Pakistan
Bachelor of Science in Computer Science 2024 – Present

- CGPA: 3.06/4 | Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Web Engineering, Artificial Intelligence

EXPERIENCE

University Project Contributor Islamabad, Pakistan
NUST SEECS — Group Projects 2024 – Present

- Developed and integrated backend services using Node.js and FastAPI, building REST API endpoints consumed by React frontends across multiple project implementations.
- Contributed to frontend development across multiple academic group projects, building responsive interfaces using React and Tailwind CSS with a focus on clean component architecture.
- Integrated RESTful APIs and managed application state within shared codebases, collaborating with teammates to deliver functional features within academic timelines.
- Worked in team settings to break down requirements, assign responsibilities, and maintain version control using Git and GitHub throughout the project lifecycle.

PROJECTS

FinMate AI | *React, FastAPI, LangChain, ChromaDB, Groq, Python* GitHub

- Built a full-stack AI banking assistant for Pakistan using FastAPI and React, featuring a RAG pipeline powered by LangChain and ChromaDB that retrieves grounded answers from banking documents with source citations.
- Integrated Groq (LLaMA3) and OpenAI as switchable LLM providers via LangChain, with local sentence-transformer embeddings, conversation history, document upload, and banking calculator tools in a dark/light mode chat UI.

Nova AI Chatbot | *React, Vite, Node.js, Google Gemini API, LangChain* Live | GitHub

- Developed a full-stack AI chatbot using Google Gemini API with a React + Vite frontend featuring real-time streaming responses and a clean chat interface.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, Java, HTML5, CSS3
AI/ML & Data: LangChain, HuggingFace Transformers, ChromaDB, Pandas, NumPy, Prompt Engineering, LLM Evaluation, RAG Pipelines
Backend: Node.js, Express.js, FastAPI, REST APIs
Frontend: React, Vite, Tailwind CSS
Databases: PostgreSQL, MySQL, MongoDB, ChromaDB
Tools & Infrastructure: Docker, Git, GitHub, Linux (command-line), Vercel